Exercise 3: Logistic regression and basis function expansion

Data

The file `pima-indians-diabetes.csv` has no header row, so the columns are named in the order given in the assignment. The target `diabetes` is converted to a factor.

Question 1: Scatterplot of glucose on age



Diabetes does not seem easy to classify with a standard logistic regression using only glucose and age. The classes overlap heavily: "Yes" and "No" points are mixed across the same age and glucose ranges. Higher glucose gives more "Yes" cases, but there is no clear separation, and a logistic regression can only produce a linear decision boundary.

Question 2: Logistic regression with glucose and age

The model is trained with $y =$ Diabetes as target, $x_1 =$ plasma glucose and $x_2 =$ age as features. Predictions for all observations use the threshold $r = 0.5$.

Probabilistic equation of the estimated model

The model assumes $y \mid x_1, x_2 \sim \text{Bernoulli}(\hat{p})$, where

$$\hat{p}(y = 1 \mid x_1, x_2) = \frac{1}{1 + \exp\left(-(-5.9124 + 0.0356\, x_1 + 0.0248\, x_2)\right)}$$

Equivalently, on the log-odds scale,

$$\log \frac{\hat{p}}{1 - \hat{p}} = -5.9124 + 0.0356\, x_1 + 0.0248\, x_2.$$

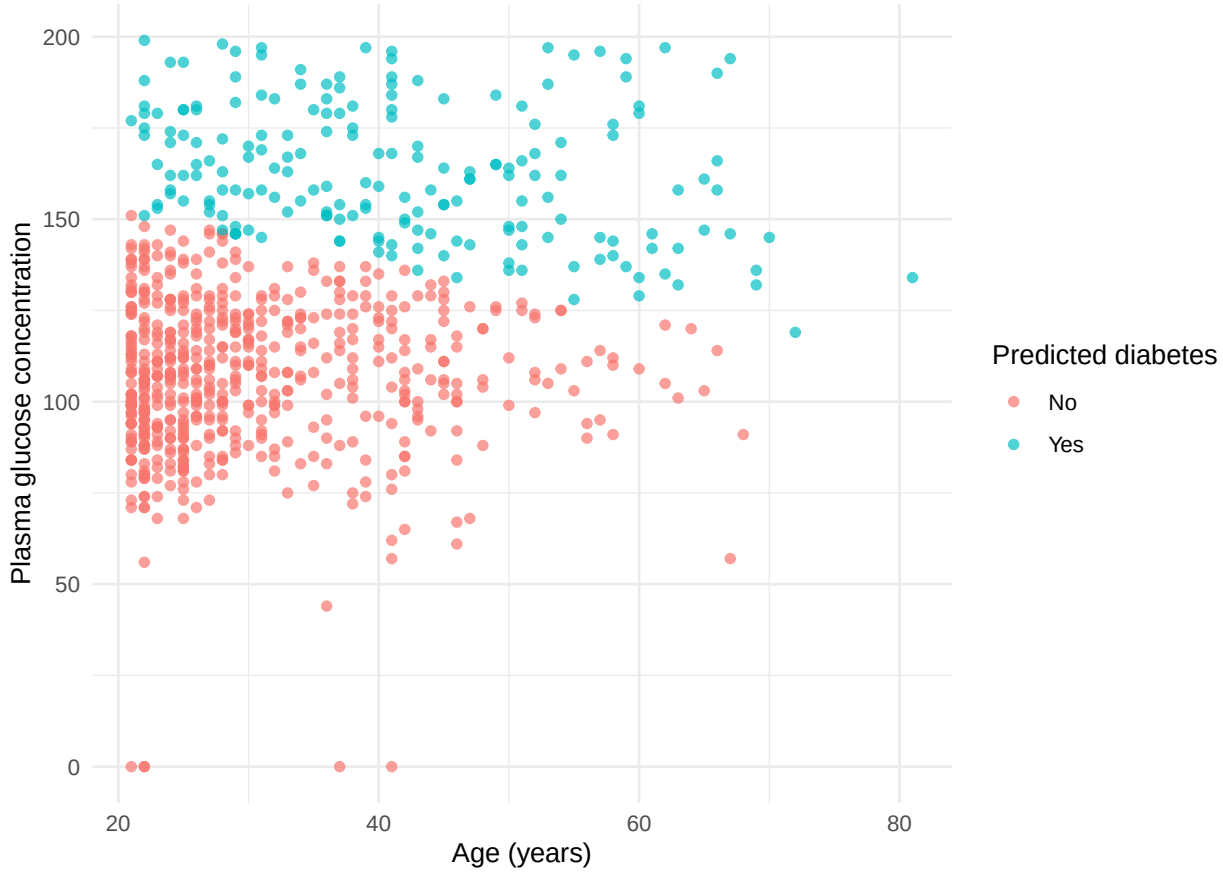
An observation is classified as Diabetes = 1 (Yes) if $\hat{p} > 0.5$, which is the same as $-5.9124 + 0.0356\, x_1 + 0.0248\, x_2 > 0$.

Training misclassification error

```
##           Actual
## Predicted  No Yes
##       No  436 138
##       Yes  64 130
```

```
## [1] 0.2630208
```

Scatterplot with predicted classes



- The training misclassification error is 0.263, so about 26.3% of the observations are classified incorrectly.
- Both glucose and age have positive coefficients, so higher glucose and higher age increase the predicted probability of diabetes. Glucose appears to have the stronger effect (larger z-value in `summary(fit)`).
- The confusion matrix shows 138 true "Yes" cases predicted as "No" and 64 true "No" cases predicted as "Yes". Thus it is more common to mistake "Yes" for "No" than the opposite.
- In the predicted-values plot the colors are separated by a straight line, while the true classes in the first plot overlap heavily. A linear decision boundary based on only glucose and age cannot separate the classes well, so the quality of the classification is only moderate. This is why a more flexible model, such as a basis function expansion, is motivated.

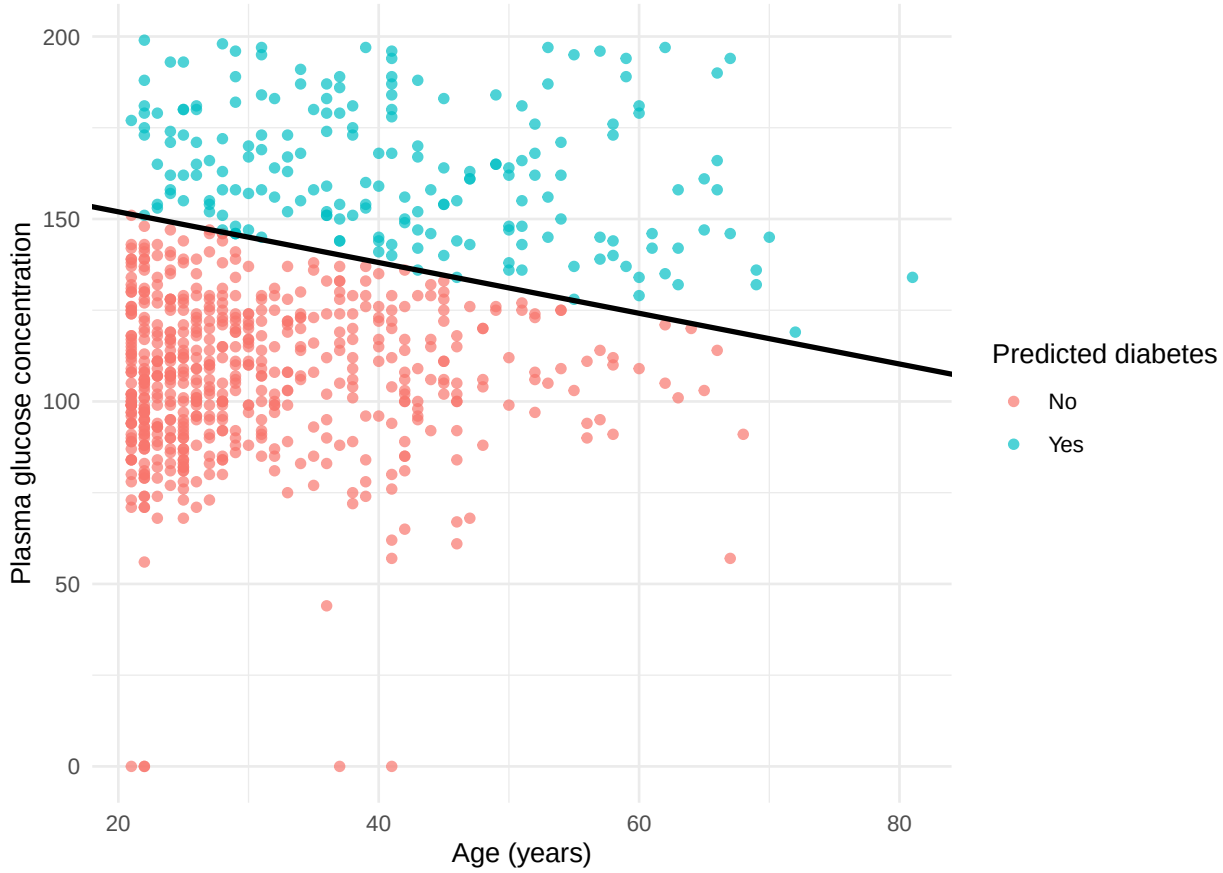
Question 3: Decision boundary

a) Equation of the decision boundary

The boundary is where $\hat{p} = 0.5$, i.e. $\hat{w}_0 + \hat{w}_1 x_1 + \hat{w}_2 x_2 = 0$. Solving for glucose ($x_1$) as a function of age ($x_2$):

$$x_1 = -\frac{\hat{w}_0}{\hat{w}_1} - \frac{\hat{w}_2}{\hat{w}_1}\, x_2 = 165.8748 + (-0.6952)\, x_2.$$

b) Boundary added to the scatterplot

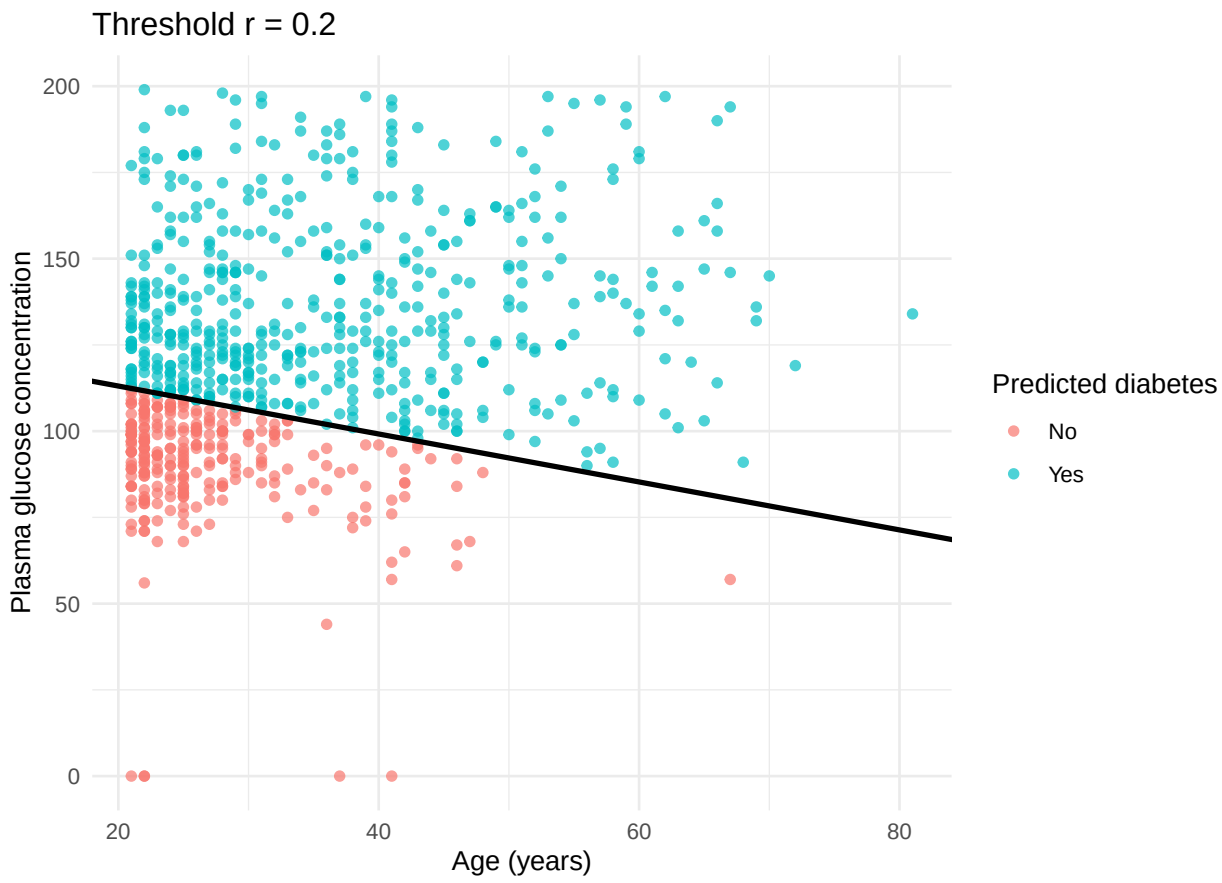


The boundary is a straight line and the true classes overlap on both sides of it. So it does not seem to capture the data distribution well, since there are many misclassified points.

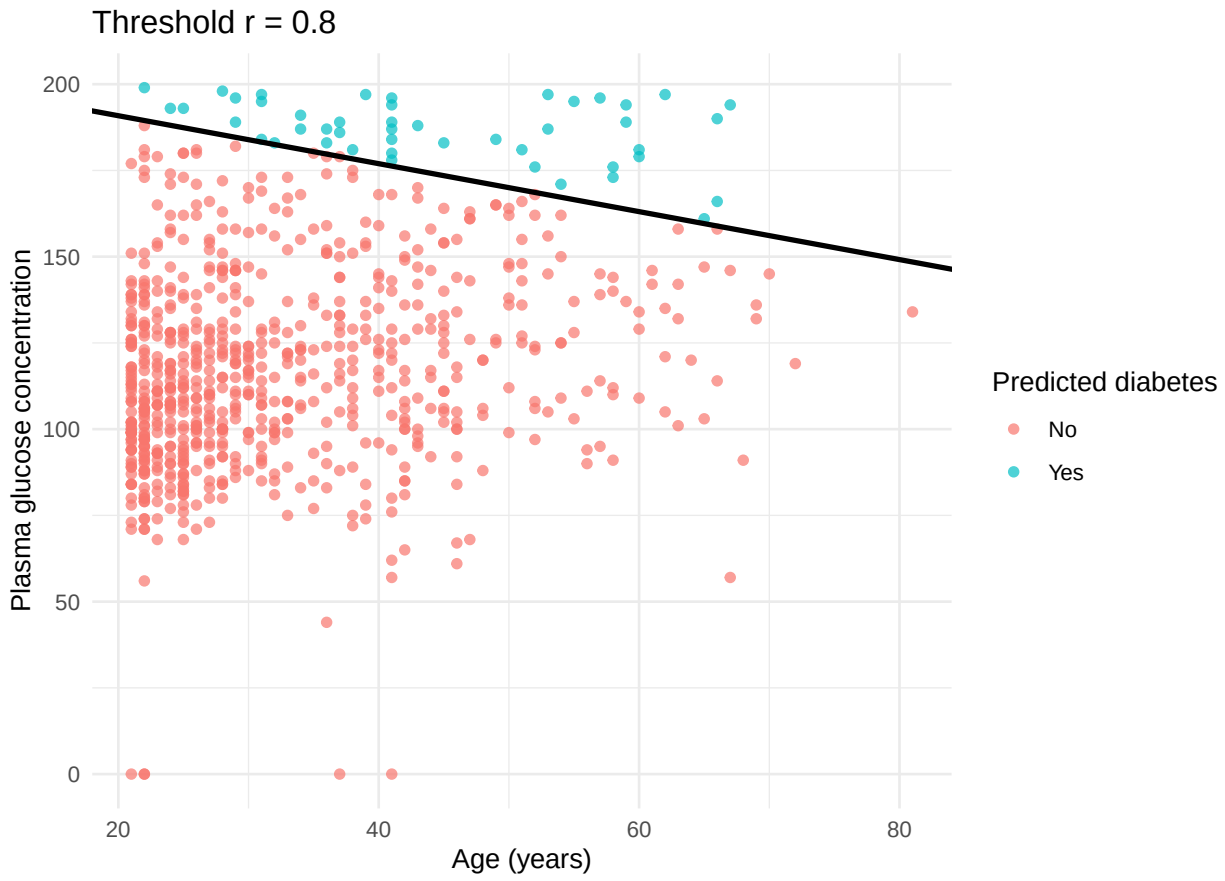
Question 4: Thresholds r = 0.2 and r = 0.8

For a general threshold r, an observation is predicted "Yes" when $\hat{p} > r$, which is equivalent to $\hat{w}_0 + \hat{w}_1 x_1 + \hat{w}_2 x_2 > \log \frac{r}{1-r}$. The boundary line therefore keeps the same slope and only its intercept changes.

```
## Threshold r = 0.2
##           Actual
## Predicted  No Yes
##       No  238  24
##       Yes 262 244
## Misclassification error: 0.3723958
```




```
## Threshold r = 0.8
##           Actual
## Predicted  No Yes
##       No  490 232
##       Yes  10  36
## Misclassification error: 0.3151042
```



The boundary line keeps the same slope and only shifts. For a smaller r, the model predicts "Yes" more easily, so more points are predicted as "Yes": 506 observations for $r = 0.2$, compared with 194 for $r = 0.5$. On the other hand, when r gets bigger there are very few "Yes" predictions (46 for $r = 0.8$), so many true "Yes" cases are missed (232 missed for $r = 0.8$, compared with 24 for $r = 0.2$).

Question 5: Basis function expansion

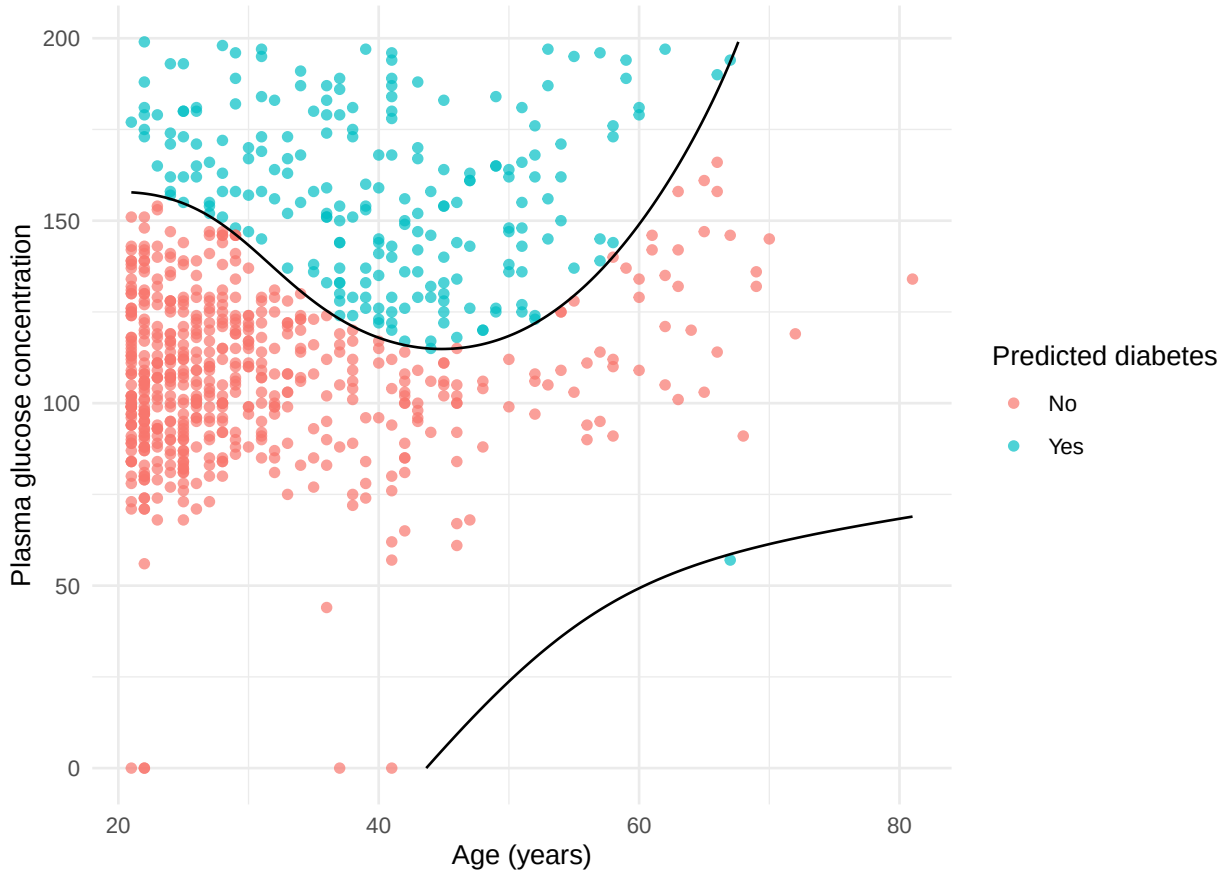
New features $z_1 = x_1^4$, $z_2 = x_1^3 x_2$, $z_3 = x_1^2 x_2^2$, $z_4 = x_1 x_2^3$, $z_5 = x_2^4$ are added to the data, and a logistic regression is trained with $x_1, x_2, z_1, \ldots, z_5$ as features.

```
##           Actual
## Predicted  No Yes
##       No  433 121
##       Yes  67 147
```

```
## [1] 0.2447917
```

Scatterplot with the curved decision boundary

The black curve is the contour where $\hat{p} = 0.5$.



The training misclassification error is 0.245 for the model with the basis expansion, compared with 0.263 for the linear model. So the accuracy improved slightly (14 fewer misclassified observations out of 768). The model finds more of the true "Yes" cases (147 instead of 130 correct), while the number of false "Yes" predictions stays about the same (67 vs 64).

The decision boundary is no longer a straight line. It is curved and non-monotonic in age: young people need a high glucose value to be predicted "Yes", the threshold is lowest around age 45-50, and then it rises steeply for older ages. This lets the model follow the data distribution more closely.

However, the improvement is small and the classes still overlap heavily, so many points remain misclassified. The flexible boundary also creates a spurious "Yes" region in the lower right, where there is almost no data, which indicates a risk of overfitting. Since this is a training error, the improvement may partly be overfitting, so the extra complexity is not clearly worth it.